

(1994). Saying that stocks are less risky over the long run is equivalent to saying that stock returns are *mean-reverting*. But the evidence for this is weak: chapter 8 shows that theory predicts and empirical evidence confirms that the amount of mean reversion in stock returns is very small.

Furthermore, if stock returns are predictable and mean-reverting, optimal asset allocation would *not* follow the one hundred minus your age rule. If stocks are mean-reverting, you want to load up on stocks when stock prices are low and expected returns are high (buy cheap), and you would reduce equity holdings when stock prices are high and expected returns are low (sell dear). Your portfolio composition would depend on the relative valuation of stocks and bonds; that is, your asset allocation would be sensitive to market conditions and your age. Luis Viceira, a finance professor at Harvard Business School, says: “It is logically inconsistent to count on reduced long-term risk while ignoring the variation in returns that produces it.”¹⁷

Pástor and Stambaugh (2012a) argue that for an investor needing to forecast the variability of stock returns (*predictive variance*, for a statistician), the risk of stocks actually increases in the long run. They show that although there is weak mean reversion in equity returns, it is swamped by the uncertainty of current and future expected returns and the large estimation risk of parameters in an econometric model used to forecast returns. Mean-reverting returns do cause stocks’ long run risk to decrease, but the other two components outweigh this effect, causing the riskiness of stocks to increase with horizon.

Protecting against Stock Underperformance in the Long Run

Zvi Bodie (1995), a pension finance expert at Boston University, demonstrates that stocks are not less risky over the long run by computing the cost of insuring against stocks underperforming bonds.

In Panel A of Figure 5.2, I graph the probability that stocks will outperform bonds as a function of horizon. All the panels in Figure 5.2 use U.S. stock and bond data from 1926 to 2012 from Ibbotson Associates.¹⁸ The probability that stocks outperform bonds rises from 58% at a one-year horizon to 86% at a thirty-year horizon. This makes sense: stock returns have a mean of 11% versus 5.7% for bonds, so stocks will eventually dominate on average. But this is not a correct statement about risk.

Panel B of Figure 5.2 plots the cost of shortfall risk. Bodie shows that this insurance cost is a put option: it pays out when the cumulated amount in stocks is less

¹⁷From Viceira (2008). Viceira’s papers and Campbell and Viceira (1999, 2001) are seminal references in dynamic portfolio allocation for individual investors. Viceira (2001) looks specifically at the case with labor income.

¹⁸The analysis assumes that stock returns are log-normally distributed and that the bond return is held constant and fixed at the mean log return of long-term government bonds. This is the Black–Scholes (1973) world.